

BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा
रंग रोगन, वार्निश और संबंधित उत्पादों के लिए नमूने और परीक्षण
भाग 6 पेंट फिल्मों पर टिकाऊपन परीक्षण
अनुभाग 2 गुणों का संरक्षण
[IS 101 (Part 6/Sec 2) का चौथा पुनरीक्षण]

**METHODS OF SAMPLING AND TEST FOR PAINTS,
VARNISHES AND RELATED PRODUCTS**

PART 6 DURABILITY TEST ON PAINT FILMS

Section 2 Keeping Properties

[Fourth Revision of IS 101 (Part 6/Sec 2)]

(ICS 87.040)

Paint, Varnishes and Related Products Sectional
Committee, CHD 20

Last Date for Comments: As per the covering letter

Paints, Varnishes and Related Products Sectional Committee, CHD 20

FOREWORD

(Formal Clause will be added later)

IS 101 "Methods of test for ready mixed paints and enamel", initially, was published as a unified standard in 1950. The standard was subsequently revised in 1961, 1964 and 1987. During the third revision, recognizing the need for clarity, the committee decided to restructure it by splitting into various parts based on type of tests. These parts included tests on liquid paints (general and physical), chemical examination, film formation, optical assessments, and mechanical tests on paint film formation. Each part was further subdivided into sections, addressing specific tests within those categories. Further, it was decided that whenever a new test method introduced, it would be integrated into the relevant part of IS 101 where it is most appropriate, ensuring that the standard remained comprehensive. Accordingly, this IS 101 (Part 6/Sec 2) was published in 1987, by superseding 31 of IS 101 : 1964 "Method of test for ready mixed paints and enamels (second revision)".

The keeping properties of paints refer to the ability of a paint to retain its original characteristics and remain in a usable condition during storage without undergoing undesirable changes. A paint with good keeping properties

should remain stable over its specified shelf life and, after stirring if necessary, should be suitable for application without any significant deterioration in performance.

This Indian Standard (Part 6) is published in several sections. The other sections of this series are:

Sec 1 Resistance to humidity under conditions of condensation

Sec 3 Moisture vapour permeability

Sec 5 Accelerated weathering test

This fourth revision has been undertaken to align the standard with current industry practices and market requirements, thereby enhancing its relevance and wider acceptance. The following major changes have been incorporated:

- a) A requirement for checking the consistency of the material after storage under specified temperature and duration conditions has been incorporated.
- b) A note on storage conditions and their simulated effects has been added for general guidance.
- c) A clause on the reporting of test results has been included to ensure uniformity in test reporting.
- d) Additionally, the various editorial corrections, and references have been updated to ensure accuracy and relevance in the revised standards.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing result of a test or analysis, shall be rounded off in accordance with IS 2: 1960 'Rules for rounding off numerical values (*revised*)'.

Draft Indian Standard

**METHODS OF SAMPLING AND TEST FOR PAINTS,
VARNISHES AND RELATED PRODUCTS**

PART 6 DURABILITY TEST ON PAINT FILMS

Section 2 Keeping Properties

(Fourth Revision)

1 SCOPE

This standard (Part 6/Sec 2) prescribes the method to determine the change in certain properties that may take place when packaged paint of either the solvent-reducible or water-reducible type is stored at a temperature above 0°C.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards:

<i>IS No.</i>	<i>Title</i>
IS 101	Methods of sampling and test for paints, varnishes and related products
Part 1	Tests on liquid paints (general and physical)
Sec 5: 2024	consistency (<i>fourth revision</i>)

3 APPARATUS

3.1 Spatula weighing (45 ± 1) g with square-ended blade, 120 mm in length and approximately 20 mm in width. It is prepared by cutting the tip from an ordinary 127 mm flexible steel laboratory spatula of the specified length.

3.2 Viscometer — A stormer viscometer with the paddle-type rotor as described in IS 101 Part 1/Sec 5.

3.3 Paint Brush — 38 mm polyester or nylon/ polyester, 13 mm thick and 63-mm length out, chisel tip.

3.4 Test Surface — A smooth surfaced paper chart coating with a suitable varnish or lacquer so as to render the surface impervious to paint vehicles.

4 PROCEDURE

4.1 Obtain two representative samples of the paint in their original, unopened containers in accordance with IS 101 (Part 1/Sec 1) (see Note 1). Where the storage test requires intermediate evaluations, obtain an additional set of duplicate samples for each scheduled examination, as mutually agreed between parties.

Examine one set of samples for the characteristics specified in **5.1** and **5.2** using the prescribed test methods. Weigh the remaining unopened samples to the nearest 1 g and store them undisturbed under the specified temperature conditions and duration agreed upon between parties (*see* Note 2). At the end of the storage period, reweigh the samples without agitation to determine any loss in mass attributable to inadequate container closure.

NOTE 1 — Test shall be carried out with the sample stored in one litre or smaller container l l.

NOTE 2 — Storage of paint for one month at (52 ± 1) °C is generally considered to simulate certain effects of storage for approximately 6 months to 1 year at (23 ± 2) °C. However, it should be noted that exposure to elevated temperatures may not merely accelerate the changes that occur under normal storage conditions. In the case of water-borne paints, for example, the growth of certain spoilage-causing microorganisms may be inhibited at 52 °C, resulting in behaviour that may differ from that observed during storage at ambient temperature.

4.2 Condition the stored samples to a temperature of (23 ± 2) °C before examination. Inspect the unopened containers for any indication of internal pressure or vacuum. Upon opening, examine the samples for the presence of skinning, settling, container corrosion, and any odour indicative of putrefaction, rancidity, or souring as per the **5.1** and **5.2**; other odours may be disregarded.

Stir the paint manually using a spatula suitable for the container for a period of 2 min (approximately 300 strokes), ensuring complete and uniform dispersion of any settled material (see Notes 3 and 4).

4.3 Immediately after stirring, determine the consistency of the paint in accordance with IS 101 (Part 1/Sec 5).

4.4 Apply the stored paint to one test panel and fresh control paint to the other. After the brushed film has completely dried, examine it for grains 0.8 mm in diameter, even larger gelatinous lumps, and streaks caused by such grains or lumps and designate as in Table 1 of **5.1**.

NOTE 3 — Avoid the use of a mechanical shaker or syringe-type measuring device that might disperse particles or lumps that would not be dispersed by hand stirring.

5 Examination of Samples

5.1 Open the container and examine the sample for skinning, internal pressure, container corrosion, and odour indicative of spoilage. Record each characteristic separately in accordance with the designations specified in Table 1.

Table 1 Designation of Changes in Skinning, Pressure, corrosion of container, odor of spoilage
(Clause 4.1 and 4.4)

Grading	Description
5	None
4	Very slight
3	Slight
2	Moderate
1	Considerable
0	Severe

5.2 Hold the can without shaking or agitation and examine the sample without the removal of supernatant vehicle. Determine the extent and character of portions of the paint that may have separated during storage by the use of spatula. Hold the spatula perpendicular to and in the centre area of the paint at a height whereby the bottom edge of the spatula is level with the top of the can. Drop the spatula from that position. Rate the conditions of sample in accordance with Table 2.

NOTE — This test shall be carried out with the sample stored in one litre or smaller container.

Table 2 Rating for Degree of Settling
(Clauses 4.2 and 4.3)

Rating	Description of Paints Condition
5	Perfect suspension. No change from the original conditions of the paint.
4	A definite feel of settling and a slight deposit brought up on a spatula. No significant resistance to sidewise movement of the spatula.
3	Definite cake of settled pigment. Spatula drops through cake to bottom of container under its own mass. Define resistance to sidewise motion of spatula. Coherent portions of cake may be removed on spatula.
2	Spatula does not fall to the bottom of container under its own mass. Difficult to move spatula through cake sidewise and slight edgewise resistance. Paint can be remixed readily to a homogeneous state.
1	When spatula has been forced through the settled layer. it is very difficult to move spatula sidewise. Definite edgewise resistance to movement of spatula. Paint can be remixed to a homogeneous state.
0	Very firm cake that cannot be reincorporated with the liquid to form a smooth paint by stirring manually.

6 REPORT

6.1 For the stored sample only, report the time of storage in days and the temperature of storage in degrees Celsius. Report the initial and final sample weight.

6.2 For both the sample tested as received and the stored sample, report the following:

6.2.1 Skinning, pressure, corrosion of the container, and odor of spoilage, each quality separately designated as follows:

5	None
4	Very slight
3	Slight
2	Moderate
1	Considerable
0	Severe

6.2.2 Degree of settling as per **4.2**;

6.2.3 Consistency as the Stormer viscosity of the paint samples (in accordance with IS 101 Part 1/Sec 5);

6.2.4 Grains, lumps, or streaks in the brushed film, designated as follows:

5	None
4	Very slight
3	Slight
2	Moderate
1	Considerable
0	Severe